CompTIA N+

M2-A1: Network Security, Maintenance & Troubleshooting Procedures

Section A: Concept Application

Q1. What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

1. CPU (Central Processing Unit):

The CPU should have sufficient processing speed, cores, and threads to run programming software, web browsers, and video conferencing applications simultaneously. It must be compatible with the motherboard socket and chipset.

2. RAM (Random Access Memory):

Sufficient RAM is required to run multiple applications smoothly. The RAM type, speed, and maximum supported capacity must be compatible with the motherboard.

3. Storage:

An SSD is recommended for faster booting, application loading, and file access. The storage type, interface, and capacity should be selected according to the system requirements and motherboard compatibility.

Conclusion:

Selecting a suitable CPU, sufficient RAM, and fast storage ensures better performance, smooth multitasking, and hardware compatibility.

Q2. Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

1. CPU Usage:

Antivirus software uses CPU resources to scan files, detect threats, and monitor system activities. During full system scans, CPU usage may increase and make the computer slower.

2. RAM Usage:

Antivirus software runs background processes that consume RAM. If available RAM becomes low, other applications may respond slowly.

3. Storage Performance:

Antivirus software reads and scans files stored on the HDD or SSD. During scanning, disk activity may increase, causing files and applications to load more slowly.

Conclusion:

Antivirus software protects the computer from malware, but heavy scanning and resource usage can temporarily reduce system performance. Scheduling scans during idle hours and keeping the antivirus updated can help reduce slowdowns.

Q3. Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

1. Check the Device Connection:

Connect the USB keyboard and printer properly. Try different USB ports to ensure the problem is not caused by a faulty port.

2. Check Device Manager:

Open Device Manager in Windows and check whether the keyboard or printer shows a yellow warning symbol or driver error.

3. Update or Reinstall Drivers:

Download the correct drivers from the manufacturer's official website. Update or reinstall the drivers according to the device and operating system requirements.

4. Check Operating System Compatibility:

Ensure that the device drivers support the installed operating system. Install available Windows updates if required.

5. Restart and Test:

Restart the computer and test the keyboard and printer. If the problem continues, run the appropriate Windows troubleshooter or check the manufacturer's support instructions.

Conclusion:

Checking connections, correcting driver issues, and verifying operating system compatibility can help restore the proper functioning of USB devices.

Q4. Explain how a UPS and surge protector help protect computer hardware and prevent data loss during power cuts and voltage fluctuations.

1. UPS (Uninterruptible Power Supply):

A UPS provides temporary battery backup when the main power supply fails. It allows users to save their work and shut down the computer safely. This helps prevent data loss and unexpected shutdowns.

2. Surge Protector:

A surge protector protects computer components from sudden voltage spikes. It helps prevent damage to sensitive hardware such as the motherboard, CPU, and storage devices.

3. Importance:

A UPS helps maintain power for a limited time during an outage, while a surge protector reduces the risk of damage caused by voltage surges.

Conclusion:

Using a UPS and surge protector helps protect computer hardware, reduce the risk of data loss, and improve system reliability during power problems.

Q5. Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

1. DNS (Domain Name System):

DNS translates domain names or hostnames into IP addresses. It allows users to access internal servers by name instead of remembering their IP addresses.

2. DHCP (Dynamic Host Configuration Protocol):

DHCP automatically assigns IP addresses and other network settings, such as the subnet mask, default gateway, and DNS server address, to devices on the network.

3. How DNS and DHCP Work Together:

DHCP provides network configuration to client devices, including the DNS server address. DNS then resolves hostnames into IP addresses so that users can access network resources by name.

4. Identifying the Problem:

Since employees can reach internal systems by IP address but not by name, the DNS service or its configuration is the most likely cause. Devices failing to obtain valid IP addresses may indicate a separate DHCP configuration or service problem.

Conclusion:

DNS and DHCP are both important for network communication. DNS resolves names into IP addresses, while DHCP automatically assigns network settings to devices. Both services should be checked to resolve the reported problems.

Q6. Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

1. Check Firewall Rules:

Review the firewall rules to identify whether legitimate traffic to the internal application server is being blocked.

2. Verify Network Connectivity:

Check the server's IP address and test connectivity using the ping command, if ICMP traffic is permitted.

3. Check Required Ports and Protocols:

Identify the ports and protocols required by the application server and verify whether the firewall allows the necessary traffic.

4. Review Firewall Logs:

Examine firewall logs to find blocked connection attempts and identify the rule responsible for blocking the traffic.

5. Correct the Firewall Configuration:

Modify the incorrect rule or create a specific allow rule for authorized traffic from the required source to the application server. Avoid disabling the entire firewall.

6. Test the Connection:

Test access to the internal application server again and confirm that the application works correctly while other security rules remain active.

Conclusion:

Checking firewall rules, logs, ports, and network connectivity helps identify and resolve traffic-blocking issues while maintaining network security.

SECTION B: PRACTICAL APPLICATION

Task 1: Workstation Assembly

Objective:

To assemble a workstation by installing the CPU, RAM, storage, and other essential components, connecting all required cables, and verifying that the system passes POST successfully.

1. Preparation of Components

Before starting the assembly, I collected all the required components and tools. These included the motherboard, CPU, RAM modules, storage device, SMPS, CPU cooler, computer case, screwdriver, and necessary cables.

I ensured that the computer was disconnected from the power supply and handled all components carefully to avoid electrostatic discharge (ESD).

2. CPU and CPU Cooler Installation

I checked the motherboard CPU socket and processor compatibility before installation. I carefully placed the CPU into the socket according to the alignment markings. After securing the CPU, I applied an appropriate amount of thermal paste if required and installed the CPU cooler correctly.

The CPU cooler helps transfer heat away from the processor and maintains a safe operating temperature.

3. RAM and Storage Installation

I checked the RAM type and motherboard compatibility before installing the memory modules. I aligned each RAM module with the DIMM slot and pressed it carefully until it was secured.

Next, I installed the compatible storage device. An SSD can improve system boot time and application loading speed compared with a traditional HDD.

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4. Motherboard Mounting and Cable Connections

I mounted the motherboard inside the computer case using the correct mounting points and screws. I then connected the main 24-pin ATX power cable and the CPU power cable from the SMPS.

I connected the storage data and power cables where required. I also connected the front-panel power switch, reset switch, indicator LEDs, and CPU cooler fan according to the motherboard manual. Before powering on, I checked that all connections were secure and that no loose screws were present inside the case.

5. Power-On Self-Test (POST) Verification

After completing the assembly, I checked the connections and powered on the workstation. During startup, the system performed the Power-On Self-Test (POST), which checks essential hardware components.

I entered the BIOS/UEFI setup to verify whether the processor, installed RAM, and storage device were detected correctly. I also checked for any hardware error messages. Successful POST and correct hardware detection indicate that the basic assembly is working properly.

Result:

The workstation assembly procedure was completed by installing the essential hardware components, connecting the required cables, and checking the system through BIOS/UEFI. Successful POST and correct hardware detection confirm that the basic assembly is functioning properly.

SECTION B: PRACTICAL APPLICATION

TASK 2: RAM MISMATCH TROUBLESHOOTING

Objective:

To identify and resolve a RAM mismatch error in BIOS/UEFI and confirm that the computer boots normally after troubleshooting.

1. Identify the RAM Mismatch Error

I started the computer and checked the BIOS/UEFI POST messages. A RAM mismatch may occur when installed memory modules have different speeds, timings, capacities, or unsupported specifications. I checked the memory information and noted any warning messages displayed during startup.

2. Check RAM Compatibility

I checked the RAM modules' capacity, speed, memory type, and supported timings. I compared these specifications with the motherboard manual to confirm that the installed RAM was supported. Differences in memory specifications can cause compatibility problems or make the memory operate at a lower supported speed.

3. Reseat or Reconfigure the RAM

I shut down the computer and disconnected the power supply before handling the RAM. I followed proper electrostatic discharge precautions, removed and reseated the memory modules correctly, and used the motherboard's recommended DIMM slots.

If necessary, I checked the BIOS/UEFI memory settings and selected supported memory settings according to the motherboard and RAM specifications. I avoided using unsupported frequencies or timings.

4. Verify the System After Troubleshooting

After reseating or reconfiguring the RAM, I powered on the computer and checked whether it completed POST successfully. I entered BIOS/UEFI and verified that the installed memory capacity was detected correctly. I then checked whether the operating system started normally without RAM-related error messages.

Result:

The RAM troubleshooting procedure involved checking POST messages, verifying memory compatibility, reseating or reconfiguring the RAM, and testing the system again. The issue is considered resolved when the BIOS/UEFI detects the memory correctly and the computer boots normally.

TASK 3: LAN TOPOLOGY BUILD

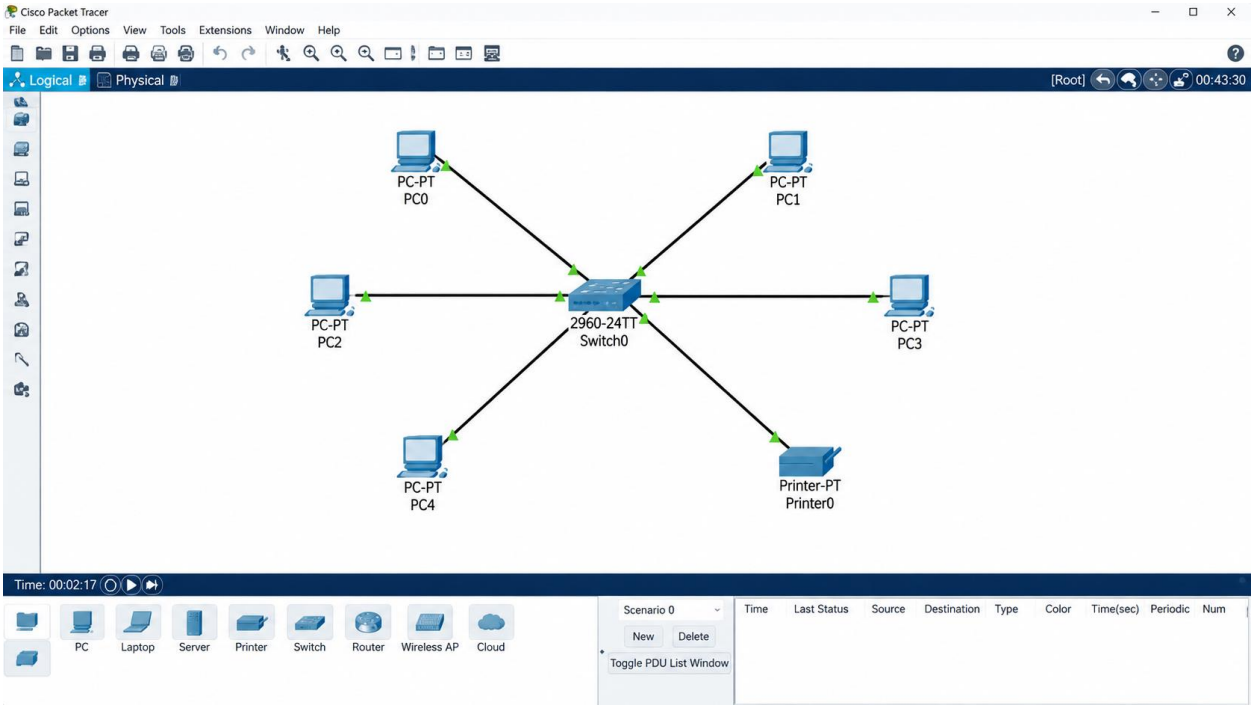
Objective:

To create a Local Area Network (LAN) in Cisco Packet Tracer containing five computer systems and one network printer, assign IP addresses from a Class C subnet, and verify connectivity using the ping command.

1. Create the Network Topology

I opened Cisco Packet Tracer and created a new network project. I added one network switch, five end systems (PCs), and one network printer. I connected all six devices to the switch using Ethernet connections.

The switch allows devices within the same LAN to communicate with each other.



Caption: Figure 1: LAN Topology with Five PCs and One Network Printer

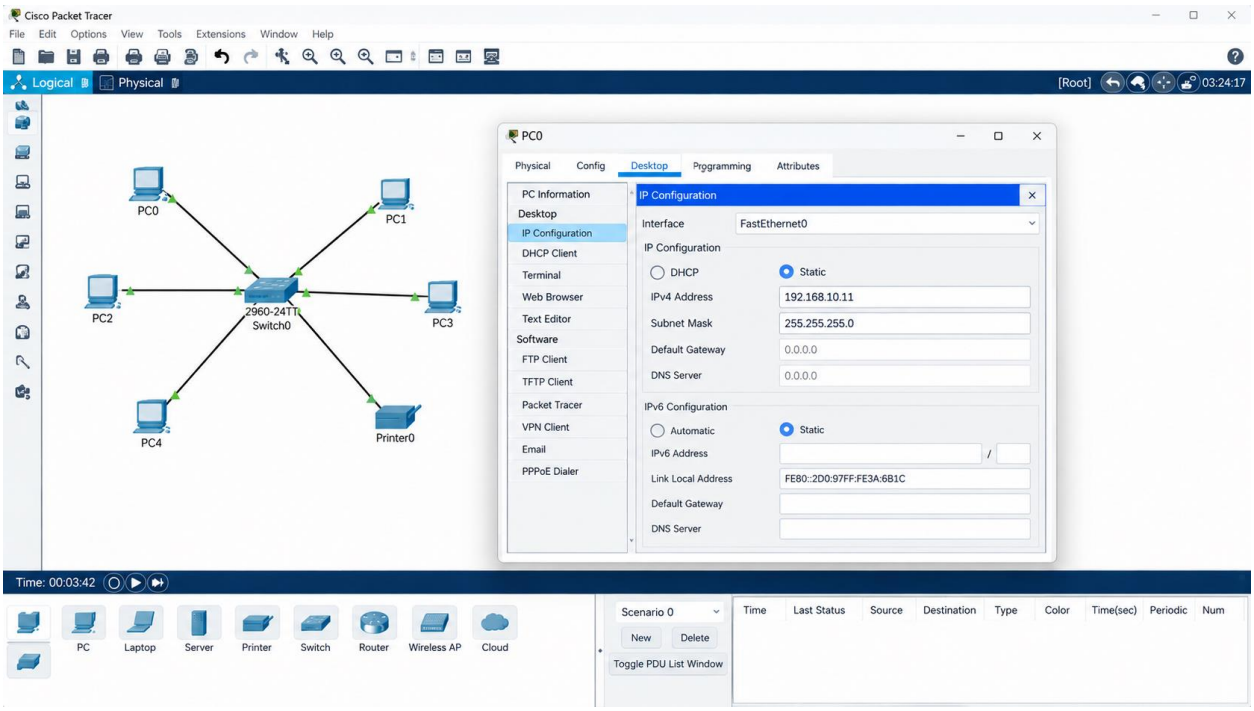
2. Assign IP Addresses

I selected the private Class C network 192.168.10.0/24 for the LAN. The subnet mask is 255.255.255.0. I assigned a unique static IP address to each PC and the network printer.

IP Addressing Table:

Device	IP Address	Subnet Mask
PC0	192.168.10.11	255.255.255.0
PC1	192.168.10.12	255.255.255.0
PC2	192.168.10.13	255.255.255.0
PC3	192.168.10.14	255.255.255.0
PC4	192.168.10.15	255.255.255.0
Printer0	192.168.10.20	255.255.255.0

All devices have addresses in the same subnet, so they can communicate directly through the switch. A default gateway is not required for communication within this same subnet. It would be needed for communication with other networks.



Caption: Figure 2: Configuring a Static IP Address on a PC

3. Configure the Network Printer

I opened the printer configuration in Cisco Packet Tracer and assigned the IP address 192.168.10.20 with the subnet mask 255.255.255.0. I checked that the printer's address did not conflict with any other device.

The printer is now part of the same LAN and can communicate with the computers through the switch.

Caption: Figure 3: Configuring the Network Printer IP Address

4. Test Connectivity Using Ping

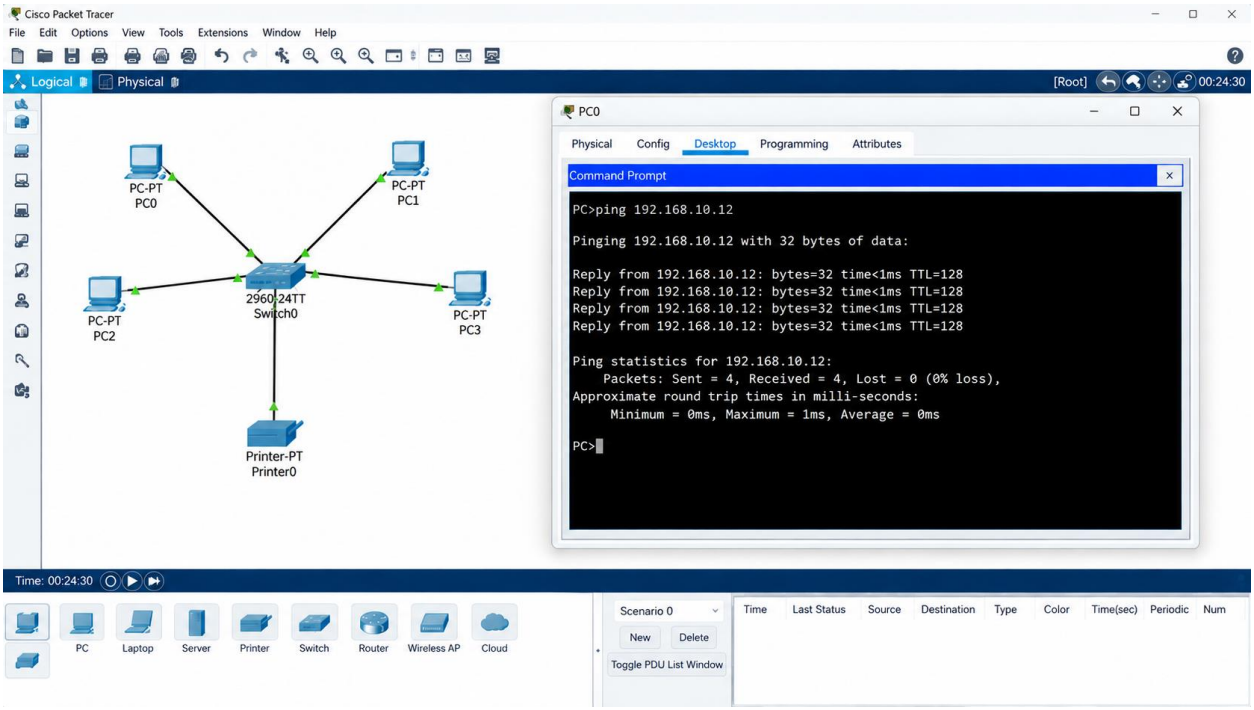
I opened the command prompt on a PC and used the ping command to test connectivity with other devices. For example, I tested communication from PC0 to PC1 and the network printer.

Commands used:

ping 192.168.10.12

ping 192.168.10.20

Successful ping replies indicate that the tested devices can communicate over the LAN. I would test all device addresses to confirm connectivity throughout the network.



4: Testing LAN Connectivity Using Ping

Result:

A LAN topology was planned with five PCs and one network printer connected through a switch. Each device was assigned a unique IP address from the 192.168.10.0/24 subnet. Successful ping tests confirm connectivity between the tested devices. All six devices should be tested in the actual Packet Tracer project before reporting the network as fully functional.

TASK 4: CLEAN OS INSTALLATION

Objective:

To perform a clean Windows/Linux operating system installation from an ISO file, manually partition the disk, optimize startup programs, and verify that the operating system boots without errors.

Note: The following procedure describes a Windows installation. A Linux installation uses a different installer and partitioning interface.

1. Prepare the Installation Media

I obtained a suitable Windows ISO and prepared a bootable USB drive using a bootable USB creation tool. Before installation, I backed up important files and checked that the computer met the operating system requirements.

I restarted the computer and opened the boot menu or BIOS/UEFI to select the bootable USB drive.

2. Boot from the Installation USB

I selected the bootable USB drive from the boot menu. After the Windows installer loaded, I selected the required language, time format, and keyboard layout. I continued to the installation screen and selected the installation option.

3. Perform Manual Disk Partitioning

I selected the custom installation option to manage the disk partitions manually. I identified the correct target disk and reviewed its existing partitions before making changes.

For a clean installation, existing partitions may need to be deleted or recreated, but this can permanently erase data. Therefore, I would verify the selected disk and back up important files before making any changes.

I created or selected the required operating system partition and allowed the installer to create any additional system partitions required by Windows. I then selected the correct target partition for installation.

4. Complete the Operating System Installation

I selected the target partition and continued the installation. The installer copied the required files and restarted the computer during the process.

After installation, I completed the initial setup, including the required user account, region, network, privacy, and other configuration settings.

5. Optimize Startup Programs and Services

After the operating system installation, I checked the installed drivers and applied the required operating system updates. I opened Task Manager and reviewed the startup applications.

I disabled unnecessary startup applications that were safe to disable. I avoided disabling essential security software, hardware drivers, or critical system services. This helps reduce unnecessary background activity and may improve startup time.

6. Verify Successful Booting

I restarted the computer and confirmed that the operating system loaded without installation errors. I checked that the keyboard, mouse, display, storage, network, and other essential devices worked correctly.

I also confirmed that the computer could restart normally and that no boot errors appeared.

Result:

The operating system installation procedure included preparing bootable media, booting from USB, manually managing disk partitions, completing installation, optimizing startup applications, and verifying successful booting. The task is complete when the operating system starts without errors and the essential hardware functions correctly.

SECTION C: MINI CAPSTONE PROJECT

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

Objective:

To identify the possible hardware and thermal causes of random workstation restarts during heavy workloads and recommend suitable troubleshooting and corrective actions.

1. Identify Three Hardware Components Most Likely to Cause Random Reboots

The three hardware components most likely to cause random restarts under heavy rendering workloads are:

A. CPU (Central Processing Unit)

The CPU performs the main processing operations. During rendering, CPU usage can become high and generate significant heat. If the processor overheats, the system may reduce performance or shut down to protect itself.

B. PSU (Power Supply Unit)

The PSU supplies electrical power to the motherboard, CPU, GPU, storage devices, and other components. If the PSU cannot provide sufficient stable power under heavy load, the computer may restart unexpectedly.

C. GPU (Graphics Processing Unit)

The GPU performs graphics processing and can consume considerable power during rendering. High GPU temperature, unstable power delivery, or a hardware fault can cause system instability.

RAM, motherboard, and other components can also cause unexpected restarts, so the three components above should be tested rather than assumed to be faulty.

2. Explain Why the Systems Pass POST in the Morning but Fail in the Afternoon

The workstations may pass POST in the morning because the components are relatively cool and the computers are not yet under heavy workloads.

During the afternoon, rendering tasks increase CPU and GPU usage and generate more heat. Poor ventilation inside desk cabinets can trap hot air and reduce cooling efficiency. As temperatures rise, the CPU or GPU may reach thermal limits.

Heavy workloads also increase power consumption. If the PSU is faulty or insufficient for the installed hardware, the system may become unstable and restart.

POST checks basic hardware operation during startup. Passing POST does not guarantee that the computer will remain stable under sustained heavy workloads.

3. Describe How to Use a Stress-Test Tool

I would use a suitable hardware monitoring and stress-testing tool to check the CPU and GPU temperatures, clock speeds, and system stability under controlled workloads.

First, I would record the temperatures and system behaviour when the workstation is idle. Then I would run a suitable CPU or GPU stress test while monitoring temperatures and system stability.

I would stop the test if temperatures reach the manufacturer's limits, the system becomes unstable, or any warning appears. I would also review system logs and hardware monitoring information for evidence of overheating or power-related instability.

If the system restarts as temperatures rise, overheating becomes a possible cause. If the system restarts under high power demand without excessive temperatures, PSU capacity, condition, and power delivery should also be investigated.

A stress test can help identify symptoms, but it does not prove a component is faulty by itself.

4. Propose Three Specific Remediation Steps

Step 1: Improve Cooling and Ventilation

Move the workstations out of poorly ventilated desk cabinets or improve airflow around them. Clean dust from the fans and heatsinks, check that all cooling fans work correctly, and verify that the CPU cooler is installed properly.

Step 2: Test Hardware and Power Delivery

Monitor CPU and GPU temperatures during controlled testing. Check the PSU's rated capacity, connectors, condition, and suitability for the installed hardware. Test suspected components using appropriate diagnostic procedures or known-good compatible parts.

Step 3: Maintain and Update the Systems

Check for relevant BIOS/UEFI updates, stable device drivers, and operating system updates. Review system logs for thermal, hardware, or unexpected shutdown events. Replace any component confirmed to be faulty and retest the workstation under normal rendering workloads.

5. Technical Justification for a PSU Upgrade

A PSU upgrade may be necessary if testing shows that the existing power supply has insufficient capacity, unstable output, a fault, or inadequate connectors for the installed CPU and GPU.

The replacement PSU should provide sufficient wattage, appropriate power connectors, good quality, and suitable protection features for the system's actual requirements.

However, a PSU upgrade should not be performed without checking the evidence. If the main cause is overheating due to poor ventilation, improving cooling may resolve the problem without replacing the PSU. The correct decision should be based on temperature measurements, power requirements, hardware tests, and system logs.

Result:

The most likely causes of random workstation restarts under heavy rendering workloads include CPU overheating, GPU overheating or instability, and PSU power-delivery problems. Improving ventilation, monitoring temperatures, testing power delivery, and reviewing system logs can help identify the root cause and reduce unexpected restarts.

SECTION D: AI-AUGMENTED LEARNING

Title: IP Addressing and Subnetting Plan for Nexus Innovations

STEP 1: BUILD WITH AI

1. Exact Prompt Use

The following prompt can be used with an AI tool to generate an IP addressing plan:

"Create an IPv4 addressing and subnetting plan for a small office network at Nexus Innovations. The network must support five computer systems and one network printer. Use a private Class C network. Specify the network address, subnet mask, usable host range, broadcast address, and a unique static IP address for each of the five systems and the printer. Reserve suitable addresses for future expansion. Explain the plan clearly and ensure that all addresses are valid and do not overlap."

2. AI-Generated Original Plan

For this report, the following is an example of an initial AI-generated plan containing an addressing error for review.

Network Address: 192.168.10.0/29

Subnet Mask: 255.255.255.248

Usable Host Range: 192.168.10.1 to 192.168.10.6

Broadcast Address: 192.168.10.7

Device Addressing:

PC0: 192.168.10.1

PC1: 192.168.10.2

PC2: 192.168.10.3

PC3: 192.168.10.4

PC4: 192.168.10.5

Printer0: 192.168.10.6

Problem Identified:

The /29 subnet provides six usable IPv4 addresses. These addresses can support the six listed devices, but the subnet has no spare usable addresses for future expansion. Therefore, it does not meet the requirement to reserve addresses for additional devices.

STEP 2: TEST AND DEBUG WITHOUT AI

3. Corrected Subnetting Plan

After reviewing the original plan, I selected a /28 subnet to provide additional usable addresses for future expansion.

Network Address: 192.168.10.0/28

Subnet Mask: 255.255.255.240

Total Addresses: 16

Usable Host Range: 192.168.10.1 to 192.168.10.14

Broadcast Address: 192.168.10.15

Corrected Device Addressing Table:

Device	IP Address	Subnet Mask
PC0	192.168.10.1	255.255.255.240

PC1	192.168.10.2	255.255.255.240
PC2	192.168.10.3	255.255.255.240
PC3	192.168.10.4	255.255.255.240
PC4	192.168.10.5	255.255.255.240
Printer0	192.168.10.6	255.255.255.240

Reserved Addresses for Future Expansion:

192.168.10.7 to 192.168.10.14

These eight usable addresses are reserved for future computers, printers, or other network devices. The network address 192.168.10.0 and broadcast address 192.168.10.15 must not be assigned to individual hosts.

4. Explanation of Changes

The original /29 subnet provided only six usable IP addresses, which were all assigned to the five computers and one printer. It did not leave any usable addresses for future expansion.

I corrected the plan by using a /28 subnet with 14 usable IP addresses. Six addresses are assigned to the existing devices, and eight addresses remain reserved for future devices.

The corrected subnet has a valid network address, subnet mask, usable host range, and broadcast address. All assigned addresses belong to the same subnet and do not overlap.

Result:

The corrected IP addressing plan supports the five computer systems and one network printer while reserving eight additional usable addresses for future expansion. This plan meets the stated requirements for the small office network.